

Product Carbon Footprint Study Report

Product Name: television production

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: television production

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 item television

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 item television.

2. System Boundary

television production in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of television production produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **353.143 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

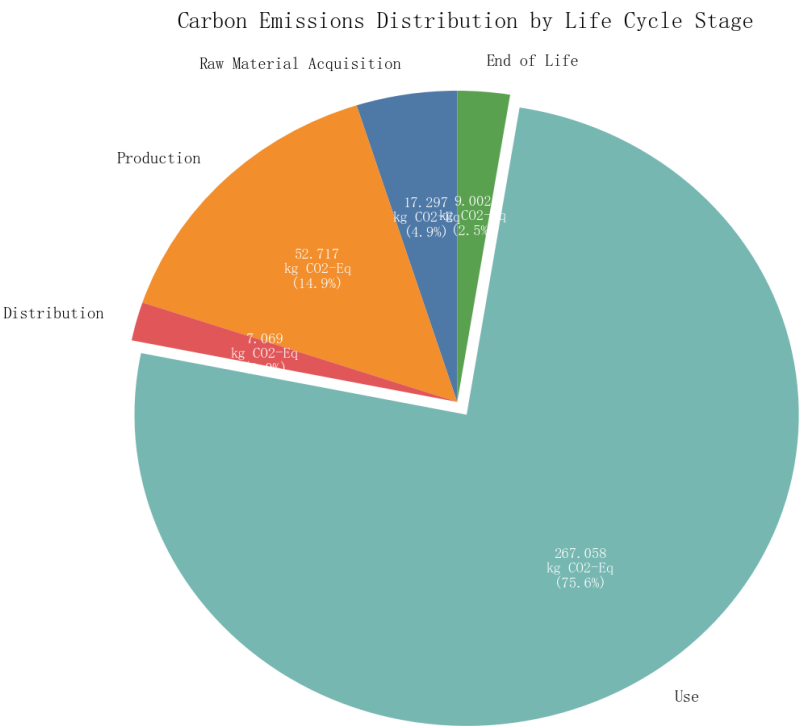


Table 1: Carbon Emissions by Life Cycle Stage

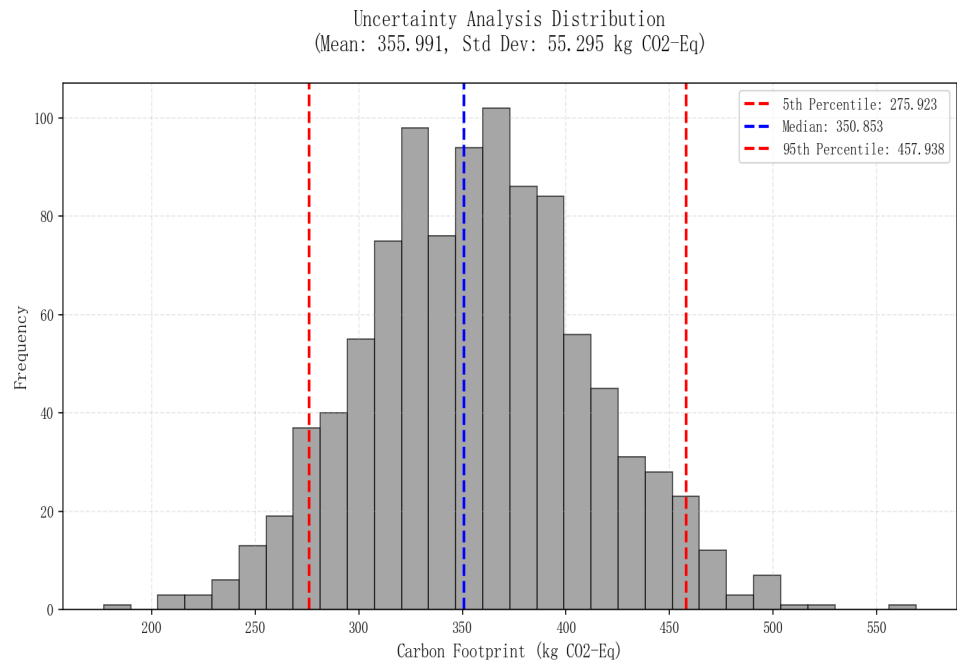
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	17.297	4.90
Production	52.717	14.93
Distribution	7.069	2.00

Use	267.058	75.62
End of Life	9.002	2.55
Total	353.143	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	355.991
Standard Deviation	55.295
Coefficient of Variation (CV)	15.53%
Median	350.853
5th Percentile	275.923
95th Percentile	457.938

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the television production is 353.143 kg CO₂-Eq, a substantial impact typical for consumer electronics where operational energy use dominates. The lifecycle stage analysis reveals a clear hotspot: the use stage accounts for 75.6% of total emissions (267.058 kg CO₂-Eq), driven by electricity consumption over the product's lifetime. In contrast, raw material extraction and production contribute 4.9% and 14.9% respectively, while distribution and end-of-life are minor at 2.0% and 2.5%. This distribution underscores that the major emission source is the use phase, likely influenced by factors such as usage patterns and grid energy mix. The top process contributions are listed as unknown, indicating a gap in process-level data that limits precise hotspot identification within each stage.

2. Statement of Assumptions and Limitations

Key assumptions in this study include a typical usage scenario with average daily viewing hours and a standard electricity grid mix, which significantly shape the use stage results. The product lifespan is likely assumed to be around 7–10 years, affecting total emissions. Limitations stem from data quality issues, as the top process contributions are all unknown, suggesting reliance on generic databases or incomplete process tracking. This reduces specificity and may mask emission sources. The uncertainty analysis shows a coefficient of variation of 15.53% and a 90% confidence interval from 275.923 to 457.938 kg CO₂-Eq, indicating moderate variability due to input assumptions. System boundary restrictions may include exclusion of capital goods or maintenance, but the cradle-to-grave scope (raw material to end-of-life) is comprehensive, though unknown processes highlight areas for improvement.

3. Suggestions for Improvement

- ****Enhance use stage efficiency:**** Implement advanced power management features, such as automatic brightness adjustment and low standby power, to reduce electricity consumption and associated emissions.
 - ****Promote renewable energy adoption:**** Encourage users to power televisions with renewable energy sources or design products with energy storage capabilities to lower grid dependency.
 - ****Optimize production processes:**** Use recycled materials and improve manufacturing efficiency for electronic components to reduce the 14.9% contribution from production.
 - ****Improve data quality:**** Conduct detailed process-level assessments to replace unknown contributions with specific data, enabling more accurate hotspot identification and targeted reduction strategies.
 - ****Extend product lifespan:**** Design for durability and repairability to delay replacement, reducing overall lifecycle impact, though this may not directly address use stage emissions.
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